



Chapter Notes: Temperature and its Measurement

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Understanding Temperature and Thermometers

Have you ever touched an ice cube and felt the chill, or placed your hands near a flame and sensed the heat? Why do some things feel cold while others feel hot?

One evening, Shruti and her younger brother Amit returned home after playing outside. Amit looked pale and tired, and their parents noticed he seemed unwell. To check if he had a fever, they used a thermometer, which confirmed he had a fever. He was advised to rest and drink plenty of water.

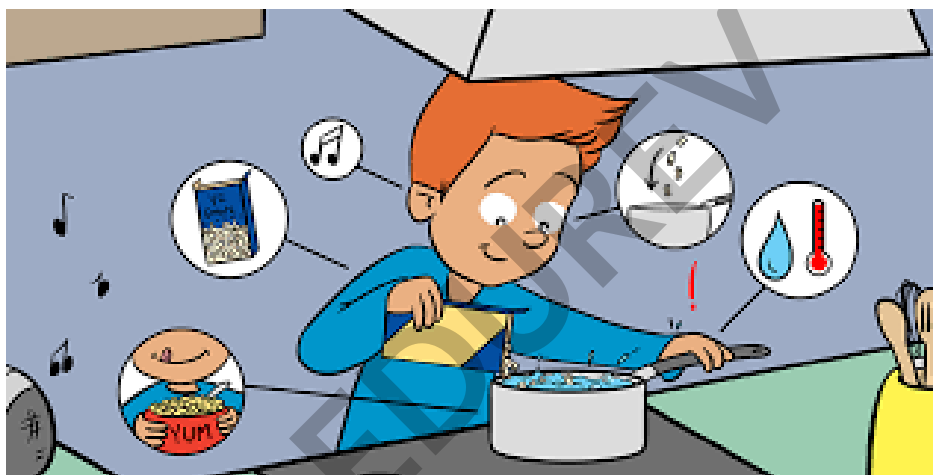


Just like Amit's parents used a thermometer to check his fever, scientists and doctors use thermometers to measure how hot or cold things are. In this chapter, we will learn about **temperature**, different kinds of **thermometers**, and how they work.

Hot or Cold?

Can we always trust our sense of touch to tell if something is hot or cold?

Not always. Our **skin** can be **tricked**. For example, a metal spoon may feel much colder than a wooden spoon even if both are at the same temperature. After being out in the sun, water at a normal temperature can feel very cold. These differences happen because materials transfer heat to our skin at different rates.



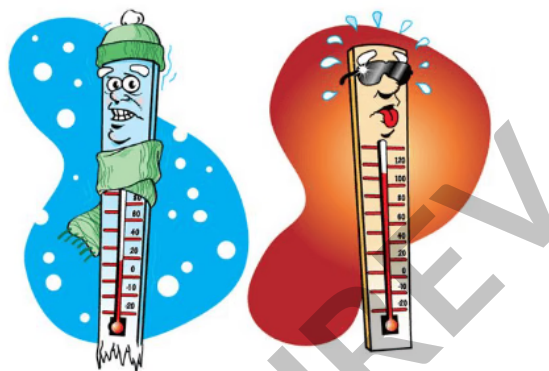
Our sense of **touch** is useful for a quick idea about temperature, but it does not give an **exact** reading. The material, its thermal conductivity, and our own body temperature affect how hot or cold something feels.



Temperature

Temperature is a measure of how hot or cold an object is. It indicates the average thermal energy of particles in the object: a higher temperature means particles move faster on average and the object has more thermal energy; a lower temperature means particles move slower and the object has less thermal energy. Heat flows from a **hotter** object to a **colder** one until both reach the same temperature (thermal equilibrium).

TEMPERATURE



Understanding temperature and its measurement is important for daily life, health, cooking, science experiments and weather forecasting.

What is a Thermometer?

A **thermometer** is an **instrument** used to measure temperature accurately. It detects changes in heat and displays the temperature on a scale.

Two main groups of thermometers are commonly used:

Clinical thermometers - Used to measure human body temperature; readings are usually in degrees Celsius ($^{\circ}\text{C}$) or degrees Fahrenheit ($^{\circ}\text{F}$).

Laboratory thermometers - Used in science experiments to measure the temperature of liquids, gases or solids.

MULTIPLE CHOICE QUESTION

Try yourself: Which type of thermometer is used to measure the temperature of the human body?

A Laboratory Thermometer

B Cooking Thermometer

C Clinical Thermometer

D Weather Thermometer

View Solution

Measuring Temperature

Clinical Thermometers



A **clinical thermometer** is designed to measure **human body** temperatures. Modern clinical thermometers are often **digital** and operate on batteries. They work by being placed in contact with a person's body (in the mouth, under the tongue, under the armpit, or in the ear) and giving a quick reading.

How do clinical thermometers work?



They measure temperature using small **heat sensors** inside the device.

Digital clinical thermometers commonly use sensors such as **thermistors** or **thermocouples** that change electrical properties with temperature.

The measured value is shown on a digital display and is powered by a battery.

Readings are usually given in degrees Celsius, shown as **°C**.

When placed correctly against the skin or in the specified body place, they provide a fast and reliable body-temperature reading.

Precautions while using a Digital Clinical Thermometer

1. Always read the **instruction manual** of the thermometer before using it.
2. Wash the **tip of the thermometer with soap and water** before and after use.
3. While washing, make sure that the **digital part (display)** does not come in contact with water.
4. Do **not hold the thermometer by its tip**. Hold it from the other end.

Temperature Scales: Celsius, Fahrenheit and Kelvin

Celsius scale - Commonly used in most countries and in school experiments; water freezes at **0 °C** and boils at **100 °C** (at standard sea-level pressure).

Fahrenheit scale - Used in some countries; an example equivalence is $37.0\text{ °C} = 98.6\text{ °F}$.

Kelvin scale - Used in science; its unit is the **kelvin** (written with a small letter) and is the SI unit for temperature.

To convert a temperature from Celsius to Kelvin use the formula:

$$\text{Celsius to Kelvin: } K = C + 273.15$$

$$\text{Kelvin to Celsius: } C = K - 273.15$$

Notes on writing units:

Scale names start with a capital letter: **Celsius, Fahrenheit, Kelvin**.

The word degree is written in small letters: degree Celsius, degree Fahrenheit.

The symbols are **°C**, **°F**, and **K**. Do not use the degree symbol with kelvin.

Place a space between the number and the unit: write 30 °C (not 30°C).

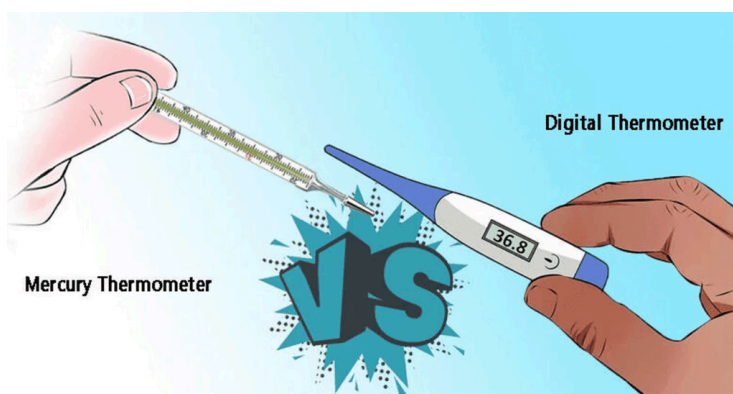
Differences between Digital and Mercury Thermometers

Mercury thermometers were once common. They use the expansion of mercury in a thin glass tube to show temperature.

Mercury is **toxic**, and broken mercury thermometers are hazardous and hard to dispose of safely.

Digital thermometers are safer because they contain no mercury. They give quick, clear readings and are easy to read.

Digital thermometers use electronic **heat sensors** for measurement and usually give readings to one or two decimal places or to the nearest 0.1 °C.



Non-contact Thermometers

Non-contact thermometers, also called **infrared thermometers**, measure temperature from a short distance without touching the person or object. During the COVID-19 pandemic, these thermometers were widely used because they reduce the risk of spreading infection.



Fever Detection Before Thermometers

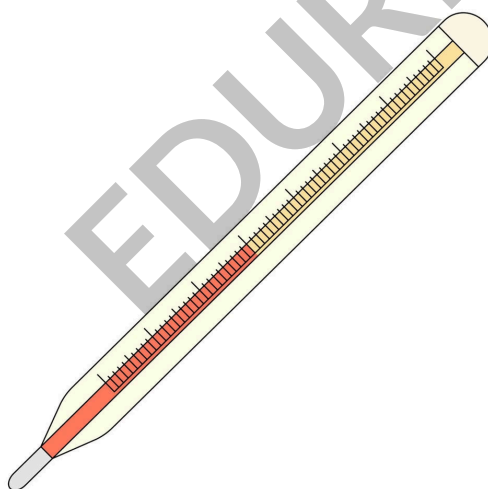
Before thermometers were widely available, people often used methods such as checking the **pulse rate** to detect fever. In ancient India, an increased pulse was linked to fever.

This method is not very accurate because many other factors-physical activity, stress, pain, or some illnesses-can also raise the pulse rate.

Therefore, a thermometer gives a more reliable measurement of body temperature.

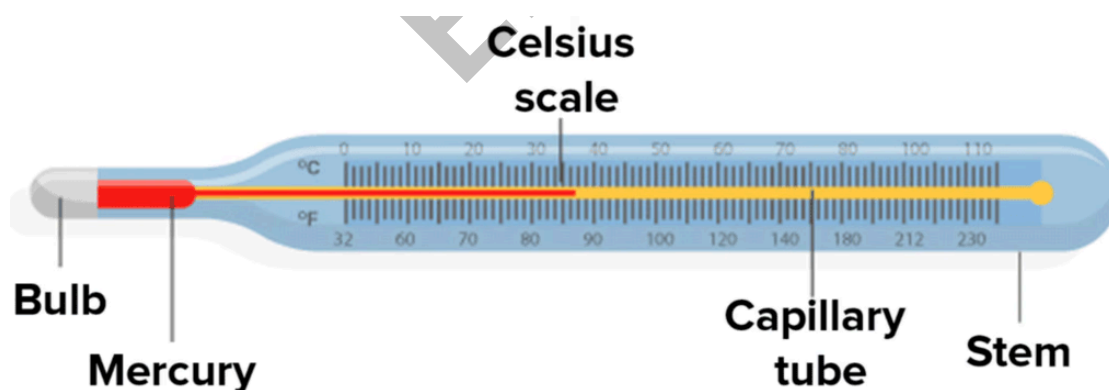
Laboratory Thermometers

A **laboratory thermometer** is used to measure temperature in science experiments. It is different from a clinical thermometer and is designed for a wider range of temperatures.



Main parts of a laboratory thermometer

1. **Capillary tube** - a long, thin glass tube sealed at one end through which the liquid rises or falls.
2. **Bulb** - a small round reservoir at the bottom that contains the liquid whose expansion is measured.
3. **Liquid column** - the thin line of liquid inside the capillary whose top marks the temperature.
4. **Celsius scale** - the marked scale alongside the tube which shows the temperature reading.
5. **Stem** - the narrow glass part containing the capillary and the scale.



Laboratory thermometers usually measure temperatures from **-10 °C to 110 °C**. They commonly contain either **alcohol** (often coloured red) or **mercury** (silver and uncoloured) as the liquid.

Reading the temperature from a laboratory thermometer

1. Hold the thermometer **upright**.
2. Look at the **top of the liquid column** in the capillary tube.
3. Read the number on the Celsius scale that **matches** the top of the liquid column.

Precautions while using a laboratory thermometer

1. Handle with care: it is made of glass and can break easily.
2. Do not shake a laboratory thermometer; shaking is not required.
3. Do not hold the thermometer by the bulb, because the heat from your hand can change the reading.
4. Do not place the thermometer in direct sunlight; this can give incorrect readings.

Fun fact!

A laboratory thermometer can be used to show the **freezing point** and **boiling point** of water: water freezes at **0 °C** and boils at **100 °C** (at standard sea-level pressure).

MULTIPLE CHOICE QUESTION

Try yourself: What is the liquid typically used in laboratory thermometers?

A Water

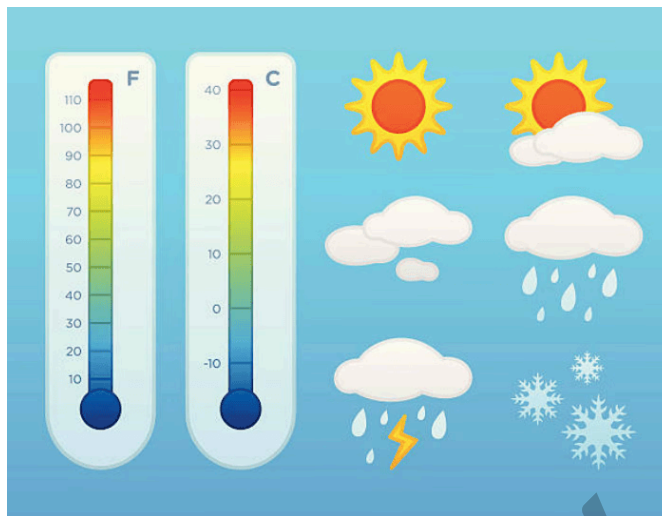
B Mercury

c Oil

D Alcohol

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Air Temperature



Weather reports in newspapers, on TV and online usually give the **highest** and **lowest** air temperatures of the day. Air temperature changes from day to day because many factors affect the weather. Temperatures usually rise toward summer and fall during winter.



Air temperature and how we measure it

Air temperature tells us how hot or cold the air is around us.

Weather stations and meteorological centres use thermometers and electronic sensors to measure air temperature.

Air temperature data, together with wind, humidity and pressure, helps meteorologists make accurate weather forecasts.

Know a scientist: Anna Mani (1918-2001) was an Indian scientist known as the 'Weather Woman of India'. She developed many weather measurement instruments and helped India reduce its dependence on imported equipment for meteorology.

MULTIPLE CHOICE QUESTION

Try yourself: What is the most common method used to measure air temperature at weather stations?

A Thermometer

B Barometer

C Hygrometer

D Anemometer

View Solution

Key Notes

1. **Temperature** measures how hot or cold something is.
2. **Thermometers** are instruments to measure temperature; common types include clinical (digital and mercury), laboratory and non-contact (infrared) thermometers.
3. **Clinical thermometers** are used for body temperature and commonly show readings in degrees Celsius ($^{\circ}\text{C}$).

4. **Temperature scales** include Celsius, Fahrenheit and Kelvin. The SI unit for temperature is the **kelvin** (K).
5. **Measurement techniques:** For accurate clinical readings, place the thermometer correctly in the prescribed body location; for laboratory measurements, read the top of the liquid column against the scale while holding the thermometer upright.
6. **Body temperature:** A typical healthy adult body temperature is about **37.0 °C** or **98.6 °F**.
Body temperature varies with age, activity and time of day.
7. **Laboratory thermometers** usually measure from **-10 °C to 110 °C** and may contain alcohol (coloured) or mercury (uncoloured).